KRIS KRUEGER

Principal Embedded Software Engineer

New York, NY · kris64aom@yahoo.com · linkedin.com/in/kristopher-krueger · https://hoplite7979.github.io/kriskrueger/

Software engineer and engineering lead with 4+ years building test automation platforms, internal tooling, and GUI-driven systems for mission-critical programs. Experienced deploying software at scale with Ansible and Jenkins, standing up complex networked test environments, and shipping features in large end-user applications. Equally comfortable writing production code and leading teams.

TECHNICAL SKILLS

Languages: Python, Tcl, Bash, JavaScript, HTML/CSS, C, MATLAB

Backend & Infra: Ansible, Jenkins, REST APIs, scripting pipelines, Git, CI/CD

Frontend & GUI: Desktop GUI development, JavaScript/HTML/CSS, end-user application features

Systems: Test asset bring-up, PC provisioning, network configuration, hardware-software integration

EXPERIENCE

Principal Embedded Software Engineer · Northrop Grumman *Aug 2026 – Present*

Baltimore, MD

- Deputy Product Owner and SME for a large GUI-based test software platform — leading sprint planning, team management, mentoring, and roadmap development
- Deploy and scale test software across production environments using Ansible and Jenkins; stand up mission-critical networked test assets end-to-end
- Design and build new features, sub-GUIs, and automation scripts in Python, Tcl, and Bash for a platform used daily by internal engineering teams
- Earned multiple individual excellence awards for consistently delivering high-quality work

Engineering Lead · Test Software Engineer · Northrop Grumman *Jul 2024 – Aug 2026*

Baltimore, MD

- Promoted to Principal Embedded Software Engineer in August 2026 after two years leading sprint planning, team management, and roadmap development as Deputy Product Owner for the platform

Systems & Integration Engineer · Northrop Grumman *Jul 2022 – Jul 2024*

Baltimore, MD

- Automated hardware board testing, analyzed flight equipment performance data, and delivered 25 RF systems and embedded computers
- Developed test software features and led cross-team integration meetings to hit program milestones

Software & Hardware Engineering Intern · AEVEX Aerospace *Jun 2021 – May 2022*

Harrisonburg, VA

- Worked on several electrical, mechanical, and software projects for aircraft hardware

Undergraduate Research Assistant · James Madison University *Jan 2019 – May 2022*

Harrisonburg, VA

- Developed MATLAB and Python tools to analyze molecular spectra; physical chemistry laboratory

PROJECTS & VENTURES

LEGO Reselling Business (BrickLink & eBay) · Founder & Operator — 4 Years

- Built and operate a profitable LEGO reselling business across BrickLink and eBay, scaling from a personal collection to an active flipping operation with thousands of listed items
- Sales Tracker — Python application that aggregates store metrics, summarizes sales history, and generates performance reports
- Price Automation Tool — Python program that cycles through thousands of inventory listings, analyzes price and sales velocity, and adjusts prices based on configurable rules; supports item exclusions, cache management, and custom filters

Low-Level C Programming

- Mandelbrot set image generator in C — fully parameterized inputs (resolution, bounds, iteration depth); hosted on GitHub
- Implemented a stack data structure in C as a learning exercise in manual memory management and pointer arithmetic

EDUCATION

B.S. Engineering — GPA 3.793 · James Madison University *May 2022*

Minors: Mathematics, Physics, Honors · President's List: 2019, 2020, 2021